

Shrinking Factor Dimension: A Reduced-Rank Approach^{*}

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Abstract

We provide a reduced-rank approach (RRA) to extract a few factors from a large set of factor proxies, and apply the extracted factors to model the cross section of expected stock returns. Empirically, we find that the RRA five-factor model outperforms the well known Fama-French five-factor model as well as the corresponding PCA, PLS and LASSO models for pricing portfolios. However, at the stock level, our RRA factor model still has large pricing errors even after adding more factors, suggesting that the representative factor proxies of our study do not have sufficient information for pricing individual stocks.

Keywords: Reduced rank, PCA, PLS, LASSO, dimension reduction

1 Introduction

Explaining why different stocks have different expected returns is one of the fundamental questions in finance and has received enormous attention over the years. Theory provides guidance for using a few factors, such as the capital asset pricing model (CAPM) of [Sharpe \(1964\)](#) which suggests the use of the market factor. But there are many factors shown to have abnormal stock returns ([Harvey, Liu, and Zhu, 2016](#); [Hou, Xue, and Zhang, 2020](#)) which are unexplained by the CAPM. Following [Cochrane \(2011\)](#), one can ask two related questions. First, how many factors do we really need? Second, given a set of well-known factors, such as the prominent five factors of [Fama and French \(FF, 2015\)](#), are there other factors that can provide incremental information for explaining the cross section of expected stock returns?

In this paper, we provide a reduced-rank approach (RRA) to address these two questions. The aim of RRA is to extract K factors from a large set of factor proxies. Since a large set of factor proxies used to explain the expected returns might suffer from the over-fitting problem. To overcome this difficulty, RRA imposes a shrinkage restriction wherein the loading matrix of all the assets has a reduced rank of K . Minimizing the mean-squared pricing error under the rank restriction will yield K factors, each of which is a linear combination of the original factor proxies, and the original regression on all the proxies (under the rank restriction) becomes a regression on these K factors. Hence, RRA shrinks the dimensionality to K . This idea is similar to the popular principal component analysis (PCA) that obtains K factors by using the first K principal components, each of which is also a linear combination of the original factor proxies. The key difference between the PCA and RRA is that RRA is designed to use its K factors to explain the returns, whereas PCA is designed to capture the K largest independent factor variances.¹

[Anderson \(1951\)](#) appears the first one to introduce RRA in statistics. [Reinsel and Velu \(1998\)](#) provide a book-level analysis on its properties and applications. In finance, [Velu and Zhou \(1999\)](#) apply a similar method to test multi-beta asset pricing models, and [Zhou \(1994\)](#) extends the

¹The scaled PCA proposed by [Huang et al. \(2022\)](#) does not apply directly to our case as it is a univariate supervised learning procedure. Instead, this paper attempts to explain multiple returns jointly.

analysis to the generalized method of moments (GMM) framework of Hansen (1982). Our paper follows the GMM set-up because it requires a much weaker distributional assumption than Reinsel and Velu (1998). Our paper differs from Zhou (1994) in two aspects. First, Zhou (1994) applies RRA to testing different types of asset pricing models, whereas we apply it to dimension reduction. Second, there is an intercept term in our case that makes the application slightly more complex.

For comparison, we also extract factors with PCA, partial least squares (PLS), and least absolute shrinkage and selection operator (LASSO), the latter two of which are originated by Wold (1966) and Tibshirani (1996), respectively, and applied to finance by Kelly and Pruitt (2013) and Feng, Giglio, and Xiu (2020), among others. In what follows, we focus on discussing empirical results from RRA because it is designed to find factors that are best to explain the basis assets (those represent the investment opportunity set) in our context, and performs the best among all the four methods.

Out of 70 major factor proxies that include the market factor, we find that, when restricted to only one factor, RRA will find a linear combination of the 70 factor proxies and this linear combination is almost identical to the market factor (with a correlation of 0.97). This indicates that the market factor is the most important one among the 70 factor proxies. Under a five-factor restriction, the RRA five-factor model outperforms the FF five-factor model for pricing FF 48 industry portfolios. This may not be surprising as FF five factors are a special case of five linear combinations out of the 70. Both sets of five factors are related, but different, as shown from their canonical correlations of 0.99, 0.89, 0.86, 0.64, and 0.13, respectively.

However, the RRA five-factor model improves little beyond the CAPM for pricing individual stocks: their root-mean-squared alpha and total adj- R^2 are close to that of the CAPM for pricing individual stocks. Moreover, the RRA ten factors improve little beyond the RRA five factors in pricing either portfolios or individual stocks. These results indicate that the 70 factor proxies do not have much incremental information beyond the RRA five factors. It is worthwhile to note that this finding and the fact that the RRA factors do not outperform existing factor models for pricing

individual stocks have little to do with the RRA methodology itself. There may be two reasons for the failure. One is that the 70 factor proxies do not capture fully the broad risk sources at the stock level. If there are some proxies that have information pricing individuals, RRA will recover them econometrically. The second reason is that the law of one price may not hold well at the stock level (than does at the portfolio level as arbitrage for mispricing will be easier here). In this case, the idiosyncratic risk is likely to be high and the existing factor proxies, even with new ones, may not be able to reduce the pricing errors at all. In terms of the factor “zoo” (Cochrane, 2011), our methodology only helps to reduce the size of the zoo, but future research is needed to explain pricing errors at the stock level.

To extract factors, both RRA and the alternative methods, except PCA, require a set of basis assets. Following the literature, we use a group of portfolios as the basis assets, which represents the opportunity set of individual assets (see, e.g., Ross, 1978; Ahn, Conrad, and Dittmar, 2009) and is a balanced panel with sufficient time series observations. The RRA factors are estimated first with the highest explanatory power on the basis assets, and then used to explain *test assets*, which are either portfolios or individual stocks. The reason for using portfolios as basis assets is that there are too many individual stocks whose unbalanced observations and dimensionality issues prevent us from using all of them. Indeed, the popular 25 size and book-to-market portfolios can be regarded as the basis assets that are used to select/design the Fama and French (1993) three factors, after 20 years, is extended to their five-factor model. In this paper, we use two sets of basis assets, FF 48 industry portfolios and 202 characteristic portfolios as in Giglio and Xiu (2021), and find similar major results, suggesting that RRA is robust to the choice of basis assets.

The RRA framework provides a simple GMM test as to whether a given set of chosen factors have zero alphas for pricing the basis assets. In addition, it allows us to impose any given level of mispricing on the basis assets. This flexibility is interesting since it allows us to price the basis assets at any desired level of accuracy, and then to examine how the pricing performance on the basis assets affects the pricing performance on test assets. Moreover, it also allows us to investigate whether the pricing ability on the basis assets impacts the out-of-sample performance for both the

basis assets and test assets.

This paper also extends RRA by pre-specifying the FF five factors as part of the true factors. Then we ask whether a few combined factors from the remaining 65 proxies can help improve the explanation of the cross section of expected pstock returns. Interestingly, we find that, for both the basis assets and individual stocks, the additional factors provide little extra explanatory power, suggesting that not much can be gained from the factor proxies beyond the FF five factors. In light of [Harvey, Liu, and Zhu \(2016\)](#), there are too many factor proxies in finance. However, our paper indicates that there are too few factor proxies that provide independent and incremental information.

[Balvers and Stivers \(2018\)](#) and [Lettau and Pelger \(2020\)](#) recently propose to extract PCA factors by imposing a pricing error restriction. Their approaches are quite interesting, but are different from ours. They do not use information on factor proxies but extract factors from basis assets directly (in their studies the basis assets are assumed to be the same as the test assets), while RRA extracts factors from factor proxies with the goal to best explain the basis assets. Hence, their approaches cannot be compared directly with RRA, as they apply to different objects.

Our paper is related to a growing number of recent studies on factors and firm characteristics. [Kelly, Pruitt, and Su \(2019\)](#) introduce instrumented PCA (IPCA) to model latent factors and time-varying loadings; [Feng, Giglio, and Xiu \(2020\)](#) and [Freyberger, Neuhierl, and Weber \(2020\)](#) **use LASSO for factor selection**; [Han et al. \(2021\)](#) introduce the use of combination forecasts and combination LASSO; and [Kozak, Nagel, and Santosh \(2020\)](#) exploit economic restrictions relating expected returns to covariance to extract factors Using Bayesian LASSO.² When comparing these studies with ours, there are two major differences. First, while most of them focus on firm characteristics, we focus on time series factors, such as the FF five factors and the alike. We ask the question of how to reduce a large number of observable time series factor proxies to a few

²Other examples are [Cong et al. \(2021\)](#), [DeMiguel et al. \(2020\)](#), [Gu, Kelly, and Xiu \(2020\)](#), [Daniel et al. \(2020\)](#), [Chen and Velikov \(2020\)](#), [Chordia, Goyal, and Saretto \(2020\)](#), [Patton and Weller \(2020\)](#), and [Avramov, Cheng, and Metzker \(2021\)](#), among others.

parsimonious factors, say from 70 to 5, which has not been resolved elsewhere.³

Second, unlike various versions of LASSO, the RRA estimates are done analytically, and the asymptotic statistical properties are known, which is a unique salient feature. Instead of removing some of factor proxies, RRA identifies the dimensionality of the factor space required to explain the basis assets. In other words, it searches sparsity in the factor space, not sparsity in the number of factor proxies. This seems particularly of interest for factor determination. In an ideal world where CAPM is true and a large panel of time series data are available for all stocks, RRA will recover the market factor, but LASSO will not because it removes many of the stocks/regressors. Theoretically, RRA is applicable regardless of whether the sparsity is in the number of factor proxies or in the space generated by them because the combination coefficient on a factor proxy can be zero if the proxy is useless.

The rest of the paper is organized as follows. Section 2 provides the RRA framework. Section 3 introduces data and key variables. Section 4 presents empirical results. Section 5 concludes.

2 Methodology

In this section, we introduce RRA and compare it with PCA and PLS. Then we discuss the performance measures and pricing error restrictions. Throughout the paper, the terms “pricing error” and “alpha” are used interchangeably.

³Barillas and Shanken (2018) pioneer a Bayesian framework, improved by Chib, Zeng, and Zhao (2020), for a similar problem. But the Bayesian approach, even ignoring linear combinations of the original factors, is computationally very difficult to implement for selecting all possible factor models from 30 given proxies due to millions of possibilities, and is almost impossible from 70 (as $2^{70} - 1$ is too large).

2.1 RRA

Following most studies, we assume that basis assets that represent the cross section of expected stock returns are governed by a latent multi-factor model,⁴

$$R_{it} = \alpha_i + \beta_{i1}f_{1t} + \cdots + \beta_{iK}f_{Kt} + u_{it}, \quad i = 1, \dots, N; t = 1, \dots, T, \quad (1)$$

where R_{it} and u_{it} are the excess and idiosyncratic returns of asset i in month t , and f_{jt} is the realization of factor j ($1 \leq j \leq K$). K and T are the number of latent factors and the number of months, respectively.

The case of wide interest is when the true number of factors, K , is typically small, say $K = 5$. The factors are assumed to be related to a number of factor proxies,

$$f_{kt} = \phi_{k1}g_{1t} + \cdots + \phi_{kL}g_{Lt}, \quad k = 1, \dots, K, \quad (2)$$

where $g_1, \dots, g_L$ are L observable variables that can be highly correlated with factor f_k . Typically, L is usually quite large, say $L = 70$. The above equation says that the few true and unknown factors are linear combinations of the L observable factor proxies. This assumption is also made for PCA and PLS, two popular dimension reduction methods, which estimate the combination coefficients differently from our approach. A detailed comparison of the three methods is provided in Sections 2.2 and 2.3, respectively.

Together with equation (2), regression (1) can be rewritten in a matrix manner as:

$$R_t = \alpha + \Theta' g_t + U_t, \quad (3)$$

where $R_t = (R_{1t}, \dots, R_{Nt})'$ is an N -vector of the basis assets' returns, $g_t = (g_{1t}, \dots, g_{Lt})'$ is an L -

⁴In this paper, we assume that the factor loadings are static and leave time-varying loadings as an open question for future research. Empirically, Section 4.1 shows that our static loading model can generate similar performance as the time-varying loading IPCA in Kelly, Pruitt, and Su (2017).

vector of the factor proxies, $U_t = (u_{1t}, \dots, u_{Nt})'$ is an N -vector of the disturbances, and Θ is an $L \times N$ matrix of the parameters. Equation (2) implies

$$\Theta = \Phi\beta, \quad (4)$$

where Φ is an $L \times K$ matrix of ϕ_{kl} and β is a $K \times N$ matrix of β_{ik} . Then it is clear that

$$H_0 : \quad \text{rank}(\Theta) \leq K. \quad (5)$$

Equation (3) tells that when the K factors are K linear combinations of the factor proxies, the rank of the regression coefficients cannot exceed K . Alternatively, if the regression coefficients have a rank of K , one must perform a dimension reduction to extract the true K factors from L proxies. Hence, estimating Φ and β is to run a reduced-rank regression in equation (3) or a regression with a rank restriction on the coefficients.

We solve below analytically by fitting the residual moment condition and, therefore, cast the problem into the framework of the generalized method of moments (GMM) (Hansen, 1982), which allows us to easily obtain the asymptotic distribution of the parameters. In terms of this paper, let $M = L + 1$ and $Z_t = (1, g_t')'$ be the instruments. The moment condition for (3) is

$$E[h_t(\alpha, \Theta)] = 0, \quad h_t(\alpha, \Theta) \equiv U_t(\alpha, \Theta) \otimes Z_t, \quad (6)$$

where $\otimes$ is the Kronecker product that makes h_t an NM -vector function of U_t and Z_t . Let h_T be the sample mean of h_t :

$$h_T(\alpha, \Theta) = \frac{1}{T} \sum_{t=1}^T h_t(\alpha, \Theta) = \text{vec}(Z'U/T), \quad (7)$$

where $Z = (Z_1', \dots, Z_T')'$, $U = (U_1', \dots, U_T')'$, and $\text{vec}(\cdot)$ is the vectorization operator. The GMM

estimator solves the following minimization problem:

$$\min_{\alpha, \Theta} Q \equiv \mathbf{h}_T(\alpha, \Theta)' W_T \mathbf{h}_T(\alpha, \Theta), \quad (8)$$

where W_T is an $NM \times NM$ positive definite weighting matrix. The resultant estimate is the GMM estimator of Hansen (1982).

Since Φ and β enter nonlinearly in the moment condition, the minimization problem (8) has to be solved numerically in general. With nonlinear restrictions, it is very difficult, if not impossible, to find the numerical solution for hundreds of parameters. However, if the weighting matrix is of the following form,

$$W_T \equiv W_1 \otimes W_2, \quad W_1 : N \times N, \quad W_2 : M \times M, \quad (9)$$

we can solve the problem analytically in two steps.

In the first step, to estimate $\Theta = \Phi\beta$, we note that Φ and β are not unique. Given any $K \times K$ non-singular matrix V , the model is the same as ΦV and $V^{-1}\beta$. That is, the estimated factors are not unique and differ only up to a rotation. However, under a suitable normalization such that the first K columns of β is an identity matrix, the estimates of Φ and β are uniquely given by

$$\hat{\Phi} = (G'PG/T^2)^{-1/2}\hat{E}, \quad \hat{\beta} = (\hat{\Phi}'G'PG\hat{\Phi})^{-1}\hat{\Phi}'G'PR, \quad (10)$$

where $R = (R'_1, \dots, R'_T)'$, $G = (g'_1, \dots, g'_T)'$, $P = P_0 - P_0\iota_T(\iota'_T P_0 \iota_T)^{-1}\iota'_T P_0$, $P_0 = ZW_2Z'$, $Z = (Z'_1, \dots, Z'_T)'$, ι_T is a T -vector of ones, and $\hat{E}$ is the $L \times K$ matrix that stacks the standardized eigenvectors ($\hat{E}'\hat{E} = I_K$) corresponding to the K largest eigenvalues of the $L \times L$ matrix:

$$(G'PG/T^2)^{-1/2'}(G'PR/T^2)W_1(G'PR/T^2)'(G'PG/T^2)^{-1/2}. \quad (11)$$

In the second step, conditional on $\hat{\Theta}$, the estimate of α can be solved analytically as

$$\hat{\alpha}' = (\iota_T' P_0 \iota_T)^{-1} \iota_T' P_0 (R - G\hat{\Theta}). \quad (12)$$

The proof is provided in Appendix A. Computationally, we can interpret the RRA estimates under the special weighting matrix (9) as running two OLS regressions (10) and (12) to fit the expected returns. However, this interpretation does not provide clear insights on the distributional properties of the estimates, which are driven by the moment conditions imposed on the original factor regression equation (3). In sum, the latent factors can be easily estimated as

$$\hat{f}_t = \hat{\Phi}' g_t. \quad (13)$$

In applications, one can standardize $\hat{f}_t$ to make it have an identity covariance matrix.

Theoretically, the estimators are asymptotically consistent, but not necessarily optimal with the minimum covariance matrix. If one is interested in improving the accuracy, it is easy to have a new estimator by using the inverse of

$$\hat{S}_T = \left(\frac{1}{T} \sum \hat{U}_t \hat{U}_t' \right) \otimes \left(\frac{1}{T} \sum Z_t Z_t' \right) = (\hat{U}' \hat{U} / T) \otimes (Z' Z / T) \quad (14)$$

as the weighting matrix, where $\hat{U}_t$ is evaluated at the previous estimator with identity weighting matrix. In our applications below, L is large relative to N or the ratio L/N is not small, the inverse of $\hat{S}_T$ is not well behaved. As a result, we simply use only the diagonal elements of $\hat{S}_T$ to obtain a consistent estimator; Section 4.1 shows that employing a more sophisticated econometric method developed by Fan, Liao, and Mincheva (2013) for estimating the covariance matrix of Z (and therefore W_2) does not change our results significantly. Of course, the best estimator can be obtained by using the optimal weighting matrix of Hansen (1982). But, as pointed out earlier, this is not feasible due to the lack of an analytical solution.

Methodologically, RRA is different from the classical reduced-rank regression (RRR) method

in two major ways (Reinsel and Velu, 1998). First, RRA is cast in a GMM framework with a weighting matrix that allows for conditional heteroscedasticity of returns while RRR does not. As a consequence, RRA has an analytical solution only for some special weighting matrices whereas RRR is solved analytically. Second, unlike most RRR studies, we impose a rank restriction on the non-intercept regression coefficients and estimate the intercepts simultaneously with other parameters (to avoid de-meaning that introduces estimation errors).

2.2 Comparison with PCA

PCA is a simple and popular dimension reduction technique and has a long history with applications in all sciences. It transforms a set of factors into independent principal components, so that the first one has the largest variance, the second one has the second largest, and so on. PCA uses a few such principal components to represent all the factors. Mathematically, it finds $\phi_k^{\text{PCA}} = (\phi_{k1}^{\text{PCA}}, \dots, \phi_{kL}^{\text{PCA}})'$ to solve successively

$$\max_{\phi_k^{\text{PCA}}} \text{Var}(g_t' \phi_k^{\text{PCA}}), \quad (15)$$

such that the later ones are orthogonal to the former. The solution is well-known. Empirically, given G , the K eigenvectors, corresponding to the first K largest eigenvalues of the $L \times L$ matrix $G'G$, are the estimates of $\phi_1^{\text{PCA}}, \dots, \phi_K^{\text{PCA}}$. Then

$$\hat{f}_{kt}^{\text{PCA}} = \hat{\phi}_{k1}^{\text{PCA}} g_{1t} + \dots + \hat{\phi}_{kL}^{\text{PCA}} g_{Lt}, \quad k = 1, \dots, K, \quad (16)$$

are the PCA factors.

By design, the first K principal components represent the best factors that explain the second moment of the L factor proxies. However, there is no guarantee that they are in any way close to the best factors that describe the average returns (i.e., the first moment) of the basis assets. Indeed, this is not surprising since no information about the basis assets is used in finding the PCA factors

except the factor proxies g_t themselves. In the worst situation, if a factor proxy has the largest variance and little ability to explain the basis assets, it will be very likely chosen as the first factor as long as it is uncorrelated with the other factors. Of course, this may not happen in real data. It just says that one needs to keep in mind that PCA is designed to explain the factor variations, rather than the average returns of the basis assets.

Note that our paper applies PCA factor proxies. Of course, PCA can be also applied to the test assets (different from the basis assets), which will be also explored in Section 4.6. There is a misconception that PCA can recover the market factor as more assets are used. That is, PCA can at most recover the projection of the market factor on the assets used, not the factor itself. A simple way to see this is to assume that there are infinite number of assets numbered with natural numbers, and one uses only odd numbered assets to perform PCA. Assume there is only one factor and it is the market factor. Then PCA recovers asymptotically only the project of the market portfolio in the space spanned by the odd numbered assets that explain perfectly these assets. In practice, PCA requires a balanced panel of asset returns, which is always a strict subset of the market as many stocks that are in the market factor in the past are delisted over time. Hence, PCA cannot recover the market factor. For both theoretical reasons and empirical performance, if one wants to use the market factor, it has to enter the factor model as an observable factor. If all factors are observable, it is model (1). If some are observable and some are latent, Section 2.7 provides a solution.

2.3 Comparison with PLS

While PCA is popular, it does not make use of information in the basis assets as it describes g_t simply by a few independent principal components of maximal variances. Recognizing this weakness, Wold (1966) introduces PLS to link factor selection to basis assets. In our context here, the objective is to search linear combinations of g_t to maximize its covariance with a linear combination of R_t . When $N = 1$, the objective is clearly to maximize the covariance of the extracted factors with R_t . When $N > 1$, it is unclear with which basis asset returns the extracted factors should

have the maximum covariance. Hence, we choose a linear combination of the basis asset returns too. Mathematically, the PLS solves

$$\max_{\psi_k^{\text{PLS}}, \phi_k^{\text{PLS}}} \text{Cov}(R'_t \psi_k^{\text{PLS}}, g'_t \phi_k^{\text{PLS}}), \quad (17)$$

where ψ_k^{PLS} and ϕ_k^{PLS} are jointly and successively solved. For our purpose, ϕ_k^{PLS} is what we need. Then the extracted factors are computed $\hat{f}_t^{\text{PLS}} = \hat{\phi}^{\text{PLS}} g_t$.

Comparing PCA and PLS with RRA, PCA ignores the information of basis assets when extracting factors. In contrast, PLS uses this information to estimate factors that have maximal covariances with the basis assets. However, in asset pricing, we are more interested in explaining the expected returns, rather than the covariances. In this sense, RRA seems particularly suitable for finance applications. It extracts factors to fit the first moment condition of the model, which is equivalent to finding the best factors to explain the expected returns of the basis assets. In addition, it is a GMM-based approach so that it is flexible in adding instrumental variables and capable of drawing inferences and testing hypotheses.

2.4 Performance measures

According to [Kelly, Pruitt, and Su \(2019\)](#), two criteria can be used to assess an asset pricing model: how it describes the cross section of expected returns and the common variation of realized returns. As such, we use two measures to assess the performance of various factor models in this paper. The first measure is root-mean-squared (RMS) alpha,

$$\text{RMS alpha} = \sqrt{\frac{1}{N} \sum_{i=1}^N \hat{\alpha}_i^2}, \quad (18)$$

which assesses to what extent the average returns are not attributable to the extracted factors.

The second measure is total adj- R^2 and defined by [Kelly, Pruitt, and Su \(2019\)](#) as

$$\text{Total adj-}R^2 = 1 - \frac{\sum_{i,t} (R_{it} - \hat{\alpha}_i - \hat{\beta}_{i1}\hat{f}_{1t} - \cdots - \hat{\beta}_{iK}\hat{f}_{Kt})^2 \times \frac{T_i-1}{T_i-K-1}}{\sum_{i,t} R_{it}^2}, \quad (19)$$

which is the fraction of return variance explained by the estimated models. Note that the summation on i is over the universe of all stocks and T_i is the number of returns of asset i . We depart from [Kelly, Pruitt, and Su \(2019\)](#) by using the total adj- R^2 so that we can compare the performance of asset pricing models with different number of factors. A subtle aspect on the total adj- R^2 , as addressed by [Kelly, Pruitt, and Su \(2019\)](#), is that the denominator is the sum of squared stock returns without demeaning. The reason for this choice is that, at the firm level, the historical mean is so noisy that using the historical mean as the benchmark will lower the bar for “good” pricing performance. Although PCA, PLS, and RRA are obtained in different ways, the above measure is straightforward to compute from regressions of excess returns on their factors, respectively.

2.5 Number of factors

In the RRA framework, given the returns of basis assets, one can conduct an analytical GMM test to explore the optimal number of factors. Although the weighting matrix given by equation (9) is not necessarily optimal under general heteroscedasticity and the usual Hansen over-identification test cannot be directly used, we can compute, based on [Zhou \(1994\)](#), an alternative GMM test as follows,

$$H_z = T (M_T \mathbf{h}_T)' V_T (M_T \mathbf{h}_T), \quad (20)$$

where V_T is an $NM \times NM$ diagonal matrix with diagonal elements $(1/v_1, \dots, 1/v_d, 0, \dots, 0)$ in which v_j is the j -th largest positive eigenvalue of the $NM \times NM$ semidefinite matrix

$$\Omega_T = [I - D_T(D_T' W_T D_T)^{-1} D_T' W_T] S_T [I - D_T(D_T' W_T D_T)^{-1} D_T' W_T]', \quad (21)$$

M_T is an $NM \times NM$ matrix, of which the i th row is the standardized eigenvector corresponding to the i th largest eigenvalue of Ω_T , D_T is the first order derivative of $\mathbf{h}_T$ with respect to α and Θ (the analytical representation of D_T is given in Appendix B, and d is the number of overidentifications. Under the rank K hypothesis, H_z is asymptotically chi-squared distributed with the degree of freedom $d = N(L + 1) - q$, where $q = N + LK + KN - K^2$ is the number of parameters.

2.6 RRA with pricing error restrictions

This section shows how the factors can be extracted conditional on a given degree of pricing error. For a fixed number of K , say $K = 5$, it is unlikely that the extracted factors have zero alphas for pricing all of the basis assets in equation (1), especially when the number of the basis assets is large. On the other hand, if the extracted factors have large alphas for some of the basis assets, there is no reason to expect them to perform well for pricing individual stocks. Hence, it is of interest to impose a certain restriction on the pricing error of the extracted factors in pricing the basis assets.

A simple way is to assume that

$$\alpha = \eta \mathbf{1}_N, \quad (22)$$

where η is a given constant and $\mathbf{1}_N$ is an N -vector of ones. For example, if $\eta = 1\%/12$, then the allowable annual pricing error is 1% across the basis assets. However, different assets have different volatilities. It is intuitive that the pricing error is likely greater for more volatile assets. Hence, it seems more sensible to consider a pricing error restriction as the following form,

$$\alpha_i = \eta \sigma_i, \quad (23)$$

where σ_i is the standard deviation of asset i . In this case, $\eta = 1\%/12$ means that the allowable annual pricing error adjusted for volatility is 1% across the assets.

With a pricing error restriction, the estimation and GMM tests are as easy as the standard case in Section 2.1. We let $P = P_0$ and replace R by subtracting α from each of its rows. However, since there are N fewer parameters, the degree of freedom of the chi-squared test becomes $N(L+1) - q_0$, where $q_0 = LK + KN - K^2$. Note that the above GMM is a joint test of a K factor model and a pricing error restriction. This is clearly more stringent than a K factor model. In other words, a greater number of factors may be needed to explain the basis assets and to satisfy the restriction.

2.7 Pre-specified factors

In the previous sections, the factors are modeled as linear combinations of factor proxies, which ignore any pre-specified factors. Based on theory and empirical studies, it is well known that the market factor is one of the most important factors (see, e.g., [Harvey and Liu, 2021](#)). Currently, the FF five factors are one set of the most studied. It is hence of interest to include the market factor or the FF five factors or some other factors as the true factors, while searching for additional factors as best linear combinations from the factor proxies.

In this case, we have the following multi-factor model for the basis assets:

$$R_{it} = \alpha_i + \beta_0' F_{0t} + \beta_{i1} f_{1t} + \cdots + \beta_{iK} f_{Kt} + \varepsilon_{it} \quad i = 1, \dots, N; \quad t = 1, \dots, T, \quad (24)$$

where F_{0t} is a K_0 -vector of pre-specified factors and β_0 is the factor loading matrix, while the rest are similar as equation (1). That is, we assume the other factors are related to the factor proxies excluding F_{0t} ,

$$f_{kt} = \phi_{k1} g_{1t} + \cdots + \phi_{kL} g_{Lt}, \quad k = 1, \dots, K, \quad (25)$$

where $g_{1t}, \dots, g_{Lt}$ are L factor proxies excluding F_{0t} .

The estimation can be done as easily as before. The only difference is to re-define t_T in equation (12) as a $T \times (1 + K_0)$ matrix of ones and pre-specified factors, and expanding α into an $N \times (1 + K_0)$ matrix to include β_0 . The degree of the GMM test is adjusted down by K_0 to reflect the pre-

specified K_0 parameters. In the case of using the FF five factors as the pre-specified factors, we simply let $K_0 = 5$ and estimate the rest from the non-FF factor proxies.

3 Data

We explore two sets of basis assets to proxy for the cross section of expected stock returns when extracting the PLS and RRA factors, as well as the LASSO factors in Section 4.7. The first set of basis assets consists of FF 48 industry portfolios, and the second set consists of 202 characteristic portfolios used by [Giglio and Xiu \(2021\)](#): 25 portfolios sorted by size and book-to-market ratio, 17 industry portfolios, 25 portfolios sorted by operating profitability and investment, 25 portfolios sorted by size and variance, 35 portfolios sorted by size and net issuance, 25 portfolios sorted by size and accruals, 25 portfolios sorted by size and beta, and 25 portfolio sorted by size and momentum. All portfolios are valued-weighted. The sample period is from January 1974 to December 2016 (516 months). To save the space, we focus on the 48 industry portfolios as basis assets in the main text and report the results with 202 characteristic portfolios in the Online Appendix.

The choice of portfolios as basis assets has good foundations. Economically, [Ross \(1978\)](#) shows that basis assets, assets that investors use in making portfolio decisions, can be reduced to a group of portfolios that dominate the opportunity set represented by individual assets, based on which an asset pricing model can be tested, rather than on a larger number of individual assets. Empirically, portfolios can efficiently reduce idiosyncratic noises in individual stock returns. According to [Ahn, Conrad, and Dittmar \(2009\)](#), a set of portfolios can be used as basis assets if stocks within portfolios are highly correlated, but have minimal correlation across portfolios. With this insight, characteristic portfolios seem satisfying the criterion.

We consider 70 factor proxies, which include the FF five factors, momentum factor, [Pastor and Stambaugh \(2003\)](#) liquidity factor, [Hou, Xue, and Zhang \(2015\)](#) ROE factor, and the

Table 1 Summary statistics of factor proxies

This table reports the average returns, CAPM alphas, and t -values of 70 factor proxies, including the FF five factors, momentum factor, [Pastor and Stambaugh \(2003\)](#) liquidity factor, [Hou, Xue, and Zhang \(2015\)](#) ROE factor, and value-weighted decile spread portfolios of 62 CAPM anomalies. The sample period is 1974:01–2016:12.

Factor proxy	Mean	α_{CAPM}	t_{CAPM}	Factor proxy	Mean	α_{CAPM}	t_{CAPM}
MKT	0.59	–	–	Ctoq	0.48	0.54	3.31
SMB	0.28	0.20	1.52	Glaq	0.45	0.58	3.28
HML	0.36	0.46	3.68	Oleq	0.66	0.87	3.81
RMW	0.30	0.38	3.79	Olaq	0.71	0.88	4.42
CMA	0.34	0.44	5.48	Claq	0.78	0.91	5.57
MOM	0.60	0.67	3.46	Oq	0.29	0.52	2.75
Liq	0.45	0.45	2.90	Olq	0.62	0.75	4.42
ROE	0.56	0.62	5.49	Kzq	0.22	0.42	2.30
Dvp	0.30	0.60	2.95	Acc	0.48	0.48	3.42
Top	0.44	0.64	3.67	Agr	0.48	0.57	3.56
Nop	0.52	0.79	4.25	Bm_ia	0.52	0.46	2.47
Ssgrow	0.33	0.46	2.94	Cashdebt	0.15	0.33	1.96
Ebp	0.43	0.42	2.17	Cfp	0.57	0.74	3.36
Ndp	0.63	0.58	2.51	Cfp_ia	0.34	0.36	2.40
Dur	0.65	0.69	3.52	Chcsho	0.55	0.70	5.02
Ndf	0.29	0.36	2.86	Chinv	0.44	0.48	3.53
Nxf	0.30	0.58	3.59	Egr	0.43	0.57	3.69
Cei	0.53	0.81	5.03	Ep	0.47	0.75	3.18
Aci	0.34	0.33	2.38	gCapx	0.36	0.45	3.03
Noa	0.53	0.57	4.03	gLtnoa	0.45	0.46	3.17
Pta	0.25	0.38	2.49	Hire	0.27	0.39	2.54
dCoa	0.22	0.30	2.05	Invest	0.51	0.58	4.21
dNco	0.25	0.24	2.34	Lgr	0.21	0.28	2.16
dNca	0.48	0.51	3.73	Orgcap	0.41	0.58	2.60
dFnl	0.33	0.39	3.35	Pchsale_Pchinvt	0.33	0.35	2.53
Cop	0.54	0.76	4.65	Pchsaleinv	0.30	0.30	2.12
F_g7	0.24	0.38	2.68	Roic	0.18	0.38	2.11
Ol	0.35	0.38	2.44	Saleinv	0.23	0.39	2.96
Rdm	0.69	0.52	2.20	Salerec	0.41	0.57	3.71
Adm	0.62	0.66	2.72	Sp	0.58	0.59	2.88
Bca	0.24	0.47	2.21	Tb	0.20	0.28	1.98
Oca_ia	0.60	0.69	5.24	Chtxq	0.53	0.46	2.42
Rnaq	0.48	0.69	3.45	Ear	0.77	0.79	5.39
Pmq	0.47	0.71	3.23	Roaq	0.57	0.78	3.78
Atoq	0.61	0.64	4.07	Roeq	0.60	0.81	3.52

value-weighted decile spread portfolios of 62 anomalies that have significant CAPM alpha (the constructions of these anomalies are detailed in the Online Appendix). Specifically, we independently replicate about 120 anomalies that are examined by [Green, Hand, and Zhang \(2017\)](#) and [Hou, Xue, and Zhang \(2020\)](#) and restrict the data to the 1974–2016 sample period. For each anomaly, we only consider the holding period of one month.⁵ Table 1 reports the average returns and CAPM alphas of the 70 factor proxies. Among these proxies, there are only two pairs of portfolios having a correlation higher than 0.90, between Ssgrow and Egr and between Roaq and Roeq. Untabulated results show that including or excluding those portfolios with correlation higher than 0.9 or 0.8 does not affect the pricing power of different factor models.

According to [Lewellen, Nagel, and Shanken \(2010\)](#) and [Daniel and Titman \(2012\)](#), the space of test assets should be high-dimensional so that a asset pricing test can provide adequate statistical power against reasonable alternative hypotheses. For this reason, we consider four sets of test assets to evaluate the pricing performance of different factor models. The first two sets of test assets are 48 industry portfolios and 202 characteristic portfolios. The third set is the universe of all domestic common stocks listed on the NYSE, Amex, and Nasdaq exchanges (i.e., stocks that have a CRSP share code of 10 or 11), and the last set is all-but-micro stocks, stocks that are larger than the NYSE 20th percentile based on market equity at the end of June each year. Both [FF \(2015\)](#) and [Hou, Xue, and Zhang \(2015\)](#) find that it is micro stocks that plague the failure of existing factor models. If a stock is delisted with missing delisting return, we follow the literature and assume a return of -30% . Finally, a stock is included in the test if it has at least 24 month returns.

⁵We do not follow [Hou, Xue, and Zhang \(2020\)](#) to treat the spread portfolios with three- and 12-month holding periods as new anomalies because they are usually and highly correlated with the one-month holding period portfolio.

4 Empirical Results

In this section, we present two main empirical results: 1) The RRA factors outperform alternative factors for pricing the 48 industry portfolios, but perform similarly for pricing individual stocks, and 2) including more factors based on existing factor proxies cannot significantly improve the pricing performance relative to the FF five factors.

4.1 In-sample performance

Table 2 reports the RMS alphas and total adj- R^2 s, defined as equations (18) and (19), of different factor models for pricing the test assets. We consider a series of FF factor models as benchmarks, where one factor refers to the market factor (i.e., CAPM), three factors to FF (1993), five factors to FF (2015), and six factors to FF (2020) (i.e., five factors plus the momentum factor), respectively. The PCA factors are the K principal components corresponding to the largest K eigenvalues of the covariance matrix of the 70 factor proxies. The PLS and RRA factors are extracted from the 70 factor proxies with the goal of maximizing the factor-return covariance and describing the average returns of the 48 industry portfolios, respectively. That is, in this table, the 48 industry portfolios are used as basis assets to represent the cross section of expected stock returns and estimate the PLS and RRA factors.

Panel A shows that, to price the 48 industry portfolios, the RRA factors perform the best. The first row shows a mixed result. In terms of explaining the first moments of the 48 industry portfolios, the FF one-factor model (i.e., CAPM) outperforms the FF multiple factor models. The RMS alpha is 0.24% with the market factor alone and 0.34% with the FF five factors. However, in terms of explaining the second moments, the multiple factor models do a better job. The total adj- R^2 s are 51.39% and 57.77% with the FF one- and five-factor models, respectively. This finding confirms Harvey and Liu (2021) that the market factor is the most important one: 1) it is the best to describe the average returns, and 2) it explains half of variations of the 48 industry portfolios,

Table 2 Performance of different factor models: In-sample

This table reports the RMS alphas and total adj- R^2 s of different K -factor models for pricing four sets of test assets: Fama-French 48 industry portfolios, 202 characteristic portfolios (Giglio and Xiu, 2021), all stocks, and all-but-micro stocks, respectively. FF refers to the Fama-French model, where 1, 3, 5, and 6 factor(s) are the market factor, FF (1993) three factors, FF (2015) five factors, and FF five factors plus the momentum factor, respectively. PCA, PLS, and RRA refer to the factors extracted from 70 factor proxies with the principal component analysis, partial least squares, and reduced-rank approach, respectively. When extracting the PLS and RRA factors, the 48 industry portfolios are used as basis assets to estimate the weights on 70 factor proxies. The 202 characteristic portfolios consist of 25 size-B/M portfolios, 17 industry portfolios, 25 operating profitability-investment portfolios, 25 size-variance portfolios, 35 size-net issuance portfolios, 25 size-accruals portfolios, 25 size-beta portfolios, and 25 size-momentum portfolios. The sample period is 1974:01–2016:12.

$K =$	RMS alpha (%)				Total adj- R^2 (%)			
	1	3	5	6	1	3	5	6
<u>Panel A: 48 industry portfolios</u>								
FF	0.24	0.29	0.34	0.31	51.39	55.57	57.77	58.34
PCA	1.00	1.06	0.88	0.99	16.74	20.49	29.92	33.14
PLS	1.03	0.54	0.33	0.24	23.42	47.19	58.97	61.11
RRA	0.23	0.20	0.18	0.18	54.60	61.11	64.75	65.38
<u>Panel B: 202 characteristic portfolios</u>								
FF	0.32	0.26	0.19	0.15	73.31	85.60	86.94	88.30
PCA	1.18	1.20	0.94	1.02	35.66	39.26	50.34	52.41
PLS	1.19	0.74	0.55	0.36	45.50	68.87	79.93	84.41
RRA	0.30	0.28	0.27	0.32	76.86	80.81	84.60	85.97
<u>Panel C: All stocks</u>								
FF	2.53	2.61	3.08	3.19	9.37	13.64	14.70	15.55
PCA	2.93	3.10	3.15	3.18	8.88	10.65	12.39	12.82
PLS	2.93	2.92	2.98	3.05	10.02	12.97	14.35	15.10
RRA	2.50	2.59	2.65	2.82	9.89	12.83	14.43	15.19
<u>Panel D: All-but-micro stocks</u>								
FF	1.78	1.85	2.06	2.09	21.07	26.98	28.39	29.25
PCA	2.21	2.36	2.29	2.33	13.63	17.53	20.81	21.72
PLS	2.25	2.13	2.02	1.95	16.35	24.23	27.96	29.47
RRA	1.73	1.80	1.78	1.83	21.74	27.45	29.79	30.61

whereas the rest factors jointly explain less than 10%. Interestingly, the momentum factor has little incremental power: it marginally reduces the RMS alpha from 0.34% with the FF five-factor model to 0.31% with the FF six-factor model and increases the total adj- R^2 by less than 1%, which

is consistent with [Fama and French \(2020\)](#) that the momentum factor describes well portfolios sorted by price momentum, but does little for describing other portfolios

The second row of Panel A shows that the PCA factors explain the 48 industry portfolios poorly. The RMS alpha is 1.01% with one factor and it is five times as large as that with the market factor, thereby suggesting that PCA extracts the first factor that has the maximum variance but a low correlation with the market factor. When the number of factors increases to 6, the RMS alpha is still 0.99% per month. The corresponding total adj- R^2 is much smaller than the FF factors, and its value is only 33.84% with six factors. Consistent with our earlier discussion in Section 2.2, the reason for the poor performance is that the PCA factors are designed to explain the variations of the factor proxies, and do not incorporate the ultimate statistical objective—describing the average returns and variations of the basis assets.

The third row of Panel A shows that the PLS factors outperform the PCA factors, and perform similarly as the FF factors when the number of factors is no less than 5. Like the first PCA factor, the first PLS factor underperforms the market factor with an RMS alpha of 1.03%. Indeed, the first PCA and PLS factors have low correlations with the market factor: 0.51 and 0.62. However, since the PLS factor model is designed to maximize the covariance between the factors and basis assets, it may be a good model for explaining the average returns, especially when the number of factors is relatively large. For example, the PLS five factors performs similarly as the FF five factors with a 0.33% RMS alpha and a 58.97% total adj- R^2 .

The last row of Panel A shows that the RRA factor model generates the smallest RMS alpha and the largest total adj- R^2 , compared with the FF, PCA, and PLS factor models. More interestingly, the first RRA factor has a 0.97 correlation with the market factor, and its total adj- R^2 is even about 3% greater than the market factor. Since the RRA factors efficiently describe the average returns, a five-factor model seems enough in explaining the 48 industry portfolios as the RMS alpha arrives at its minimum at $K = 5$.

To obtain more insights as to why RRA differs from PCA and PLS, Figure 1 plots the weights

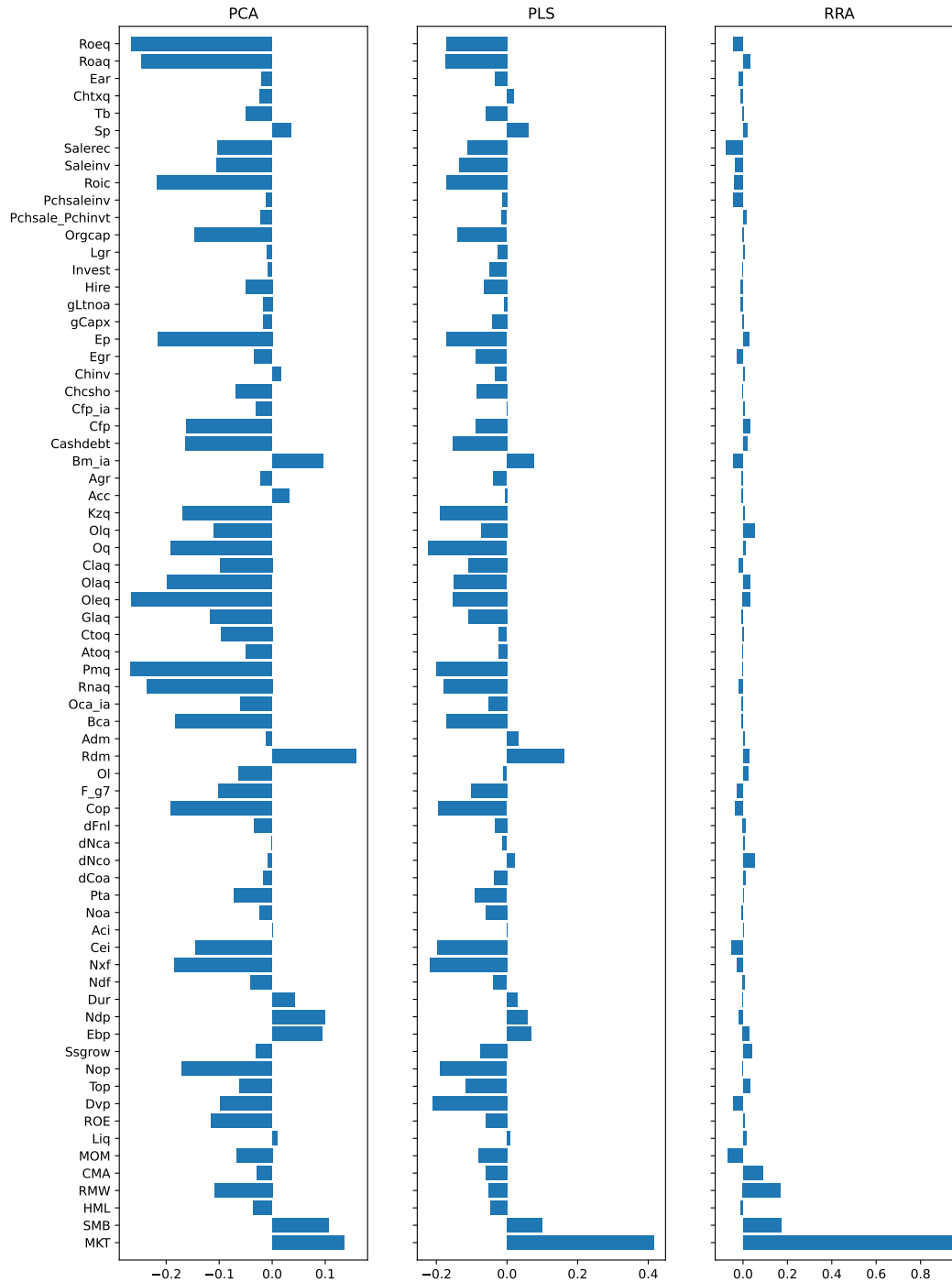


Figure 1. Weights of the first PCA, PLS, and RRA factors

This figure plots the weights of the first PCA, PLS, and RRA factors on the 70 factor proxies, respectively. For comparison, the weights are normalized to have an L_2 -norm of 1. The sample period is 1974:1–2016:12.

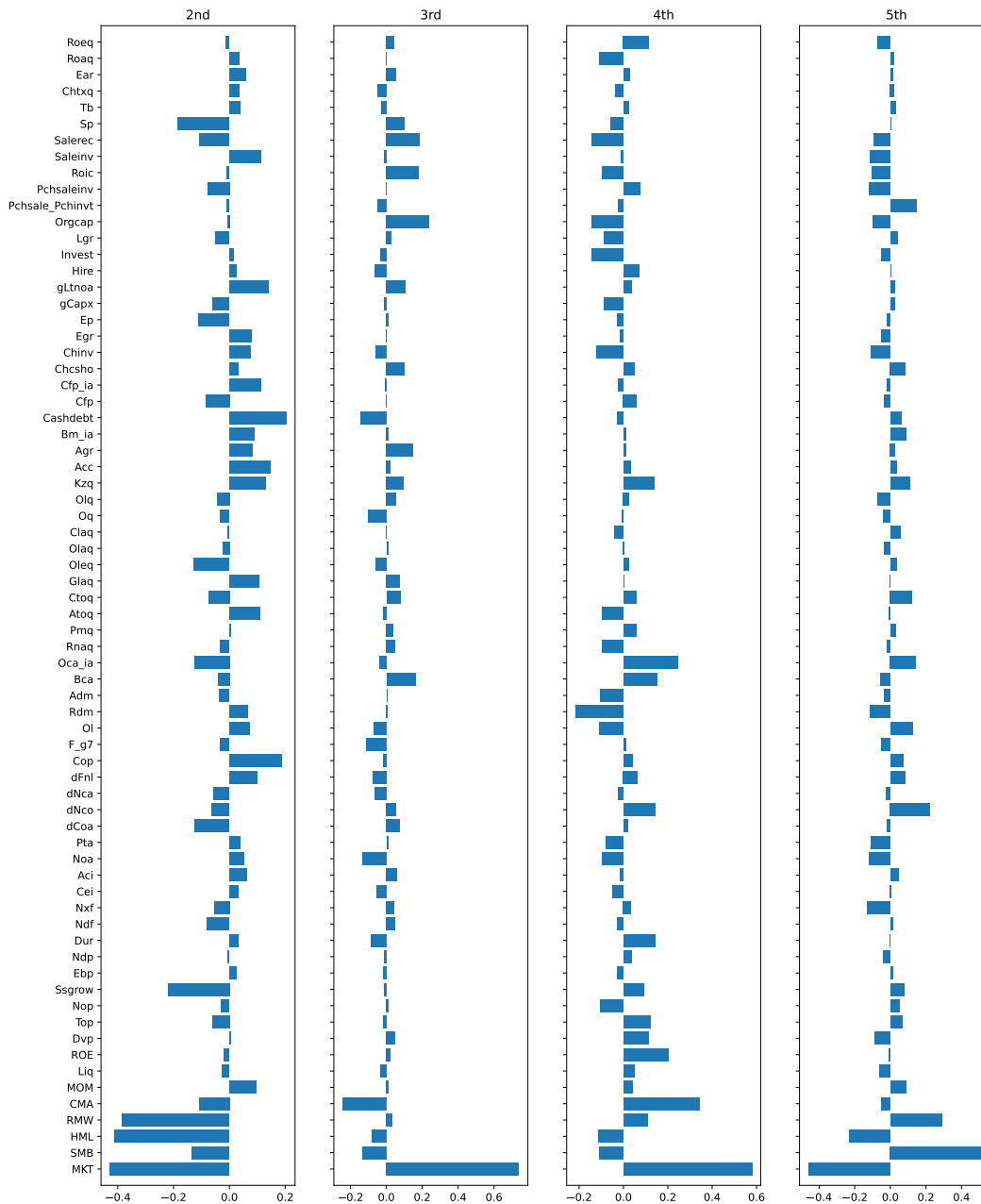


Figure 2. Weights of the second-fifth RRA factors

This figure plots the weights of the second-fifth RRA factors on the 70 factor proxies, respectively. For comparison, the weights are normalized to have an L_2 -norm of 1. The sample period is 1974:01–2016:12.

of the first PCA, PLS, and RRA factors on the 70 factor proxies, respectively. For comparison, we normalize the weights of each factor to have an L_2 -norm of 1. The loadings of the three factors on the market factor are 0.14, 0.42, and 0.94, respectively, thereby explaining why the market

factor has a low correlation with the PCA and PLS factors but a high correlation with the RRA factor. Thus, RRA performs in the way as it is designed to do so. Figure 2 plots the weights on the second through fifth RRA factors, respectively. Ignoring the weight on the market factor, the second RRA factor tilts toward FF's value and profitability factors (HML and RMW), the third toward Eisfeldt and Papanikolaou's (2013) organizational capital (Orgcap) and FF's investment factor (CMA), the fourth toward FF's investment factor (CMA) and Eisfeldt and Papanikolaou's (2013) industry-adjusted organizational capital to assets (Oca_{ia}), and the fifth toward FF's size factor (SMB), respectively. To have a closer look at the relationship between the FF five factors and the RRA five factors, we find that their canonical correlations are 0.99, 0.89, 0.86, 0.64, and 0.13. This result shows that the RRA five factors are related to, but also different from, the FF five factors. Figure 3 plots the absolute alphas of the four factor models in describing the 48 industry portfolios and clearly shows that the RRA absolute alphas are almost always smaller than the PCA and PLS's.

In Panel A of Table 2, the test assets are the same as basis assets, and the RRA factors are estimated with basis assets. This makes the RRA factors easier to outperform the FF factors. Panels B through D apply the extracted factors in Panel A to price three sets of different assets. To some extent, the resultant results can be regarded as the out-of-sample performances of the PLS and RRA factors. Also, they mitigate the critics of Lewellen, Nagel, and Shanken (2010) and Daniel and Titman (2012) that low-dimensional test assets or treating test assets the same as basis assets cannot provide adequate statistical power against reasonable alternative hypotheses.

Panel B considers the 202 characteristic portfolios in Giglio and Xiu (2021) as test assets. Since most of the portfolios are finer sorts of the FF five factors, the pricing power should tilt toward the FF model and the result confirms this inference. The RMS alpha and total adj- R^2 are 0.32% and 73.31% with the market factor and 0.19% and 86.94% with the FF five factors, suggesting that the non-market factors are helpful for pricing the 202 characteristic portfolios. The PCA factors perform the worst as Panel A. The PLS factors do not do a fair job either and its six-factor RMS alpha is even larger than that with the market factor alone (0.36%). The RRA factors are of interest.



Figure 3. Absolute alphas of 48 industry portfolios

This figure plots the absolute alphas of 48 industry portfolios with the FF, PCA, PLS, and RRA five-factor models, respectively. The sample period is 1974:01–2016:12.

They significantly outperform the PCA and PLS factors and similarly as the FF factors. When $K = 5$, the RMS alpha and total adj- R^2 are 0.27% and 84.60%, respectively. Thus, even the test assets are different from the basis assets, the RRA factors continue to do a fair job.

Panel C provides the first main finding of this paper. When the test assets are individual stocks, all the factor models perform similarly when they are restricted to have the same number of factors. More surprisingly, when the number of factors increases, the RMS alpha does not increase in general, but even slightly increases. In contrast, the total adj- R^2 increases significantly from $K = 1$ to 5, and marginally to 6. These results suggest that a model with more than five factors provides little incremental power in explaining the average returns and variations of individual stock returns relative to a five-factor model. Panel D repeats Panel C but excludes microcap stocks. The purpose for this test is that, as argued by both FF (2015) and Hou, Xue, and Zhang (2015), it is microcap stocks that plague extant factor models. The results of this panel show that while the pricing power of all the factor models improve, their patterns are similar to Panel C and the factors are far from perfect to explain individual stocks. The RMS alphas are about 2% per month and the total adj- R^2 's are less than 30%.

Our results in Panels C and D are comparable with Kelly, Pruitt, and Su (2019). In their paper, they consider 12,813 individual stocks over the 1962–2014 sample period, and the total R^2 with their IPCA five-factor model and the FF five-factor model are 18.6% and 21.90%, respectively. In comparison, due to different data filters, we cover 19,255 stocks in Panel C and 6,888 stocks in Panel D over the 1974–2016 period. With our sample, the total adj- R^2 of the FF five factors is 14.70% for all stocks and 28.39% for all-but-micro stocks. More importantly, the corresponding RRA five factors generate a total adj- R^2 of 14.43% for all stocks and 29.79% for all-but-micro stocks. This result suggests that, even we impose a stronger assumption (i.e., constant risk exposure), the RRA factors perform similarly, or even better than, as the Kelly, Pruitt, and Su (2019) factors with time-varying risk exposures.

Since both PLS and RRA use information in the basis assets in extracting the factors, one

Table 3 Extracting PLS and RRA factors using alternative basis assets

This table uses 202 characteristic portfolios as basis assets for estimating the weights on the 70 factor proxies when extracting the PLS and RRA factors. Reported are the root-mean-squared alphas and total adj- R^2 s in pricing four sets of test assets. The sample period is 1974:01–2016:12.

$K =$	Root-mean-squared alpha (%)				Total adj- R^2 (%)			
	1	3	5	6	1	3	5	6
Panel A: 48 industry portfolios								
PLS	1.00	0.38	0.29	0.25	21.60	47.32	57.99	59.77
RRA	0.25	0.28	0.27	0.25	51.14	58.00	60.49	62.32
Panel B: 202 characteristic portfolios								
PLS	1.17	0.51	0.26	0.26	43.42	71.33	83.68	84.49
RRA	0.30	0.23	0.16	0.15	79.71	86.93	88.97	89.46
Panel C: All stocks								
PLS	2.91	2.88	3.06	3.10	9.89	13.04	14.74	15.18
RRA	2.45	2.70	2.96	2.97	11.56	14.37	15.91	16.26
Panel D: All-but-micro stocks								
PLS	2.22	2.09	1.97	1.98	15.83	24.34	28.44	29.22
RRA	1.73	1.87	1.90	1.92	23.54	28.02	29.88	30.62

Table 4 Performance of RRA factors: An alternative estimation

This table reports the RMS alphas and total adj- R^2 s of RRA K -factor models in explaining four sets of test assets, where the weighting matrix W_2 in (9) is estimated with the principal orthogonal complement thresholding method of [Fan, Liao, and Mincheva \(2013\)](#). The 48 industry portfolios are used as basis assets. The sample period is 1974:01–2016:12.

$K =$	RMS alpha (%)				Total adj- R^2 (%)			
	1	3	5	6	1	3	5	6
48 industry portfolios	0.23	0.20	0.19	0.18	54.63	61.38	64.83	65.54
202 characteristic portfoliosp	0.29	0.27	0.26	0.32	75.86	81.01	83.86	85.59
All stocks	2.51	2.59	2.64	2.91	9.53	12.82	14.21	15.05
All-but-micro stocks	1.74	1.80	1.78	1.85	21.19	27.38	29.74	30.53

natural question is whether their pricing performance is sensitive to the choice of the basis assets. To answer this question, we use the 202 characteristic portfolios as basis assets to estimate the factors and report the results in Table 3. Overall, the PLS and RRA factors are insensitive to the choice of basis assets, and the magnitudes and patterns are quantitatively similar as Table 2.

Due to high dimensionality, we assume the weighting matrix to be diagonal in extracting the RRA factors. This assumption is perhaps not realistic because the covariance matrix of the factor proxies, as shown in equation (14), is unlikely diagonal. To deal with this potential issue, we follow Fan, Liao, and Mincheva (2013) by assuming the factor proxies are governed by latent factors and estimating their covariance matrix with the principal orthogonal complement thresholding (POET) method. The results are reported in Table 4 and quantitatively similar as Table 3, which suggests that the weighting matrix is not a determinant for the pricing power of the RRA factors.

In sum, we conclude that when the test assets are portfolios, one model may have greater pricing power than the other; but when the test assets are the universe of individual stocks, the FF, PCA, PLS, and RRA models perform similarly, although the FF are based on a couple of factors whereas the PCA, PLS, and RRA are based on 70 factor proxies. Therefore, based on existing factor proxies, little is gained beyond using the FF five factors in describing the cross section of expected stock returns.

4.2 Out-of-sample performance

In the previous section, we examine the performances of the PCA, PLS, and RRA factors based on the full sample of data. One natural question is how the performances change over time if the factor weights are constructed ex ante. To answer this question, we use the first 30-year data to estimate the weights, and assign the weights to the rest 13-year data to estimate the PCA, PLS, and RRA factors (the results are similar if we use the first 20-year data for weight estimation). For each individual asset, we employ a 60-month rolling window approach, requiring at least 24 observations, to calculate the out-of-sample RMS alphas and total adj- R^2 s. As a result, the out-of-sample evaluation period is January 2004 to December 2016.

Table 5 reports the out-of-sample results. Specifically, in Panel A when the test assets are the basis assets, the RMS alphas of the FF factors are 50% larger than their in-sample values. The PCA factors still perform the worst but slightly better than the in-sample case. The PLS factors

Table 5 Performance of different factor models: Out-of-sample

This table reports the out-of-sample RMS alphas and total adj- R^2 s of different K -factor models in pricing four sets of test assets. Different from Table 2, we use the first 30-year data to estimate the weights of the PCA, PLS, and RRA factors and the rest for out-of-sample evaluation. For each individual asset, we employ a 60-month rolling window approach, requiring at least 24 observations, to calculate the out-of-sample performance. As such, the out-of-sample evaluation period is 2006:01–2016:12.

$K =$	Root-mean-squared alpha (%)				Total adj- R^2 (%)			
	1	3	5	6	1	3	5	6
Panel A: 48 industry portfolios								
FF	0.46	0.53	0.57	0.48	53.66	53.72	51.20	51.82
PCA	0.98	0.97	0.85	0.86	30.26	31.73	34.50	32.24
PLS	0.93	0.56	0.41	0.40	33.62	44.50	54.29	54.04
RRA	0.46	0.35	0.32	0.34	55.36	55.40	56.65	55.89
Panel B: 202 characteristic portfolios								
FF	0.29	0.30	0.24	0.22	78.22	85.04	84.73	86.27
PCA	1.05	1.02	0.84	0.86	49.71	49.88	52.50	50.06
PLS	0.97	0.53	0.28	0.26	52.90	65.81	75.59	80.21
RRA	0.29	0.25	0.24	0.26	80.65	81.01	83.25	83.90
Panel C: All stocks								
FF	2.86	2.97	3.07	3.14	6.46	−0.41	−9.02	−16.39
PCA	2.94	2.96	2.95	2.99	4.66	−1.74	−9.83	−15.29
PLS	2.92	2.92	2.94	2.97	4.98	−0.50	−7.34	−11.99
RRA	2.83	2.86	2.90	2.98	7.37	1.22	−7.82	−12.91
Panel D: All-but-micro stocks								
FF	2.01	2.03	2.09	2.05	20.01	15.94	8.20	2.05
PCA	2.15	2.14	2.10	2.18	11.92	7.71	2.27	−2.94
PLS	2.11	2.01	2.01	2.03	13.12	12.45	9.37	5.87
RRA	2.00	1.99	2.01	2.06	21.50	17.25	10.85	6.84

underform the FF factors with $K \leq 3$ and outperform with $K > 3$. The RRA factors continue to outperform the FF factors, and again, achieve the minimum RMS alpha and the maximum total adj- R^2 at $K = 5$. Overall, the weights for extracting the PCA, PLS, and RRA factors are relatively stable over time, consistent with [Kelly, Pruitt, and Su \(2019\)](#), [Lettau and Pelger \(2020\)](#), [Kozak, Nagel, and Santosh \(2020\)](#), and [Freyberger, Neuhierl, and Weber \(2020\)](#), among others.

In Panels B, C, and D, the FF and RRA factors perform similarly. They outperform the PCA

and PLS in terms of both RMS alpha and total adj- R^2 . When pricing all or all-but-micro stocks, the RRA factors do not improve very much over the FF factors, consistent with the early in-sample analysis that existing other factor proxies do not provide much incremental pricing power.

4.3 Number of factors

We perform the GMM test of Section 2.5 to explore how many factors we need to explain the cross section of expected stock returns. Specifically, to explain the average returns of the 202 characteristics portfolios, we examine the suitable number of factors. Untabulated results show that the chi-square statistic H_z in equation (20) is large and rejects a model with up to ten factors. Similar results are also obtained if we use the original Hansen's (1982) GMM test by solving problem (8) for the parameters with aid of our analytical solution as the starting point for numerical iterations.

Our result that the GMM test rejects a model with as many as 10 factors is consistent with Kozak, Nagel, and Santosh (2020) who show that it is difficult to find a sparse stochastic discount factor with a few factors to explain 50 anomalies examined in their paper. However, the GMM result only says that there are more than ten factors that are needed to fit well the model moment conditions. This is a model assessment from the statistical perspective. Economically, though, our empirical results show that the number of factors beyond five improves little on the pricing errors, implying that the number of factors may be taken as a number around five for practical purposes. Note that the factors are extracted from the 70 factor proxies, so the fruitful way to reduce pricing error is not to add more existing factor proxies into the model, but to find new factor proxies that contain incremental information.

4.4 Performance of RRA factors with pricing error restrictions

This section explores how a pricing error restriction affects the pricing power of the RRA factors. Without loss of generality, we consider the pricing error constraint as equation (23) with four scenarios: $\eta = 0, 0.5/12, 1/12$, and $1.5/12$, suggesting that the pricing error ranges from zero to $1.5/12$ times of the standard deviation. Conceptually, if a K -factor model prices the basis assets well, a zero alpha restriction does not affect the pricing performance, and in contrast, a non-zero alpha restriction will reduce the pricing performance.

Table 6 reports the results. Note that over our sample period, the monthly average volatility of the 48 industry portfolios is 6.5%. Hence, while a restriction of $\eta = 0$ implies that we do not allow any pricing error when extracting the RRA factors, a restriction of $\eta = 1/12$ suggests that we allow a pricing error as large as 1.64% per year for each asset when extracting the RRA factors.

Table 6 makes two observations. First, imposing a zero alpha restriction does not dramatically affect the pricing performance of the RRA factors. For example, in Panel A when the test assets are the 48 industry portfolios, in the case of no pricing error ($\eta = 0$), the three- and five-factor RMS alphas are 0.19% and 0.16%, which are quantitatively similar to the case without restriction (0.20% and 0.18%). Similarly, in Panel C when the test assets are all individual stocks, the corresponding values are 2.57% and 2.64%, which are also extremely close to the values without restriction: 2.59% and 2.65%. This pattern also applies to the total adj- R^2 .

Second, a non-zero alpha restriction has a stronger effect on RMS alpha than the total adj- R^2 for pricing portfolios. For example, when $\eta = 1/12$, the five-factor RMS alpha and total adj- R^2 are 0.54% and 82.94% for pricing the 202 characteristic portfolios. In this case, without restriction, the corresponding values are 0.27% and 84.60%. However, for pricing individual stocks, a non-zero alpha restriction has a negligible effect on both the RMS alpha and total adj- R^2 . In Panel C when $\eta = 1/12$, the five-factor RMS alpha and adj- R^2 are 2.68% and 14.20%, in contrast to 2.65% and 14.43% without imposing the restriction. Therefore, this section shows that a pricing error restriction may affect the RMS alpha for pricing portfolios, but has a negligible effect on

Table 6 Performance of RRA factors estimated with a pricing error restriction

This table reports the root-mean-squared alphas and total adj- R^2 s of RRA K factors in pricing four sets of test assets. The factors are extracted by imposing a pricing error restriction as $\alpha_i = \eta \sigma_i$, where σ_i is asset i 's volatility. The sample period is 1974:01–2016:12.

$K =$	Root-mean-squared alpha (%)				Total adj- R^2 (%)			
	1	3	5	6	1	3	5	6
Panel A: 48 industry portfolios								
$\eta = 0$	0.23	0.19	0.16	0.16	54.60	60.98	64.73	65.32
$\eta = 0.5/12$	0.30	0.26	0.24	0.23	54.33	60.70	64.45	65.03
$\eta = 1/12$	0.44	0.41	0.38	0.38	53.63	60.00	63.72	64.30
$\eta = 1.5/12$	0.59	0.57	0.55	0.54	52.49	58.92	62.56	63.14
Panel B: 202 characteristic portfolios								
$\eta = 0$	0.30	0.29	0.29	0.32	76.70	80.92	84.60	85.84
$\eta = 0.5/12$	0.39	0.40	0.40	0.43	76.30	80.50	84.06	85.26
$\eta = 1/12$	0.53	0.55	0.54	0.57	75.27	79.44	82.94	84.11
$\eta = 1.5/12$	0.69	0.71	0.70	0.73	73.79	77.77	81.23	82.40
Panel C: All stocks								
$\eta = 0$	2.50	2.57	2.64	2.77	9.83	12.87	14.43	15.15
$\eta = 0.5/12$	2.50	2.58	2.65	2.79	9.84	12.82	14.35	15.06
$\eta = 1/12$	2.52	2.60	2.68	2.83	9.77	12.71	14.20	14.92
$\eta = 1.5/12$	2.57	2.64	2.72	2.89	9.67	12.51	14.02	14.72
Panel D: All-but-micro stocks								
$\eta = 0$	1.74	1.78	1.76	1.79	21.68	27.38	29.77	30.61
$\eta = 0.5/12$	1.73	1.79	1.77	1.82	21.57	27.23	29.59	30.42
$\eta = 1/12$	1.76	1.83	1.82	1.87	21.26	26.94	29.25	30.08
$\eta = 1.5/12$	1.83	1.89	1.89	1.95	20.89	26.50	28.75	29.58

individual stocks.

4.5 Performance of pre-specified factors

In this section, we answer the second question in this paper: given the prominent FF five factors, are there other factors that provide incremental information for explaining the cross section of expected stock returns?

Table 7 presents the results when the pre-specified factors are the FF five factors and the rest are

the PCA, PLS, and RRA factors, in which the non-FF factors are extracted from the factor proxies excluding the FF five factors. In Panel A when the test assets are the 48 industry portfolios, the RMS alpha of the FF five factors is 0.34%. When augmenting the FF five factors with PCA, PLS, or RRA factors, the PCA and PLS factors do not help much, the RRA factors are helpful when three or five more are included. For example, an eight-factor model can produce an RMS alpha of 0.21% with RRA factors. However, when the test assets are the 202 characteristic portfolios, the improvement is negligible because the FF five factors have already done a good job.

In Panels C and D when the test assets are individual stocks, the RMS alpha is even higher when including PCA, PLS, or RRA factor. For example, it is 3.08% with the FF five factors alone and 3.18% when augmenting the FF five factors with one RRA factor. This fact is true for both PCA and PLS factors. The total adj- R^2 increases slightly but it is not economically sizeable. Therefore, non-FF factors could not provide much incremental power for pricing portfolios and individual stocks relative to the FF five factors.

Theoretically, a factor model outperforms another if its factors have a higher Sharpe ratio, representing better the investment opportunity set (see, e.g., [Gibbons, Ross, and Shanken \(1989\)](#) and [Barillas and Shanken \(2018\)](#)). Empirically, this may not be true because of estimation errors, especially when we conduct relative model comparison in this paper. To see this is the case, we consider the [Daniel et al. \(2020\)](#) five factors, which have hedged the unpriced components in the FF five factors and have a much higher Sharpe ratio. Unreported results show that the [Daniel et al. \(2020\)](#) five factors do not improve the performance of the FF five factors in explaining our four sets of test assets, and the performance is even slightly worse. For example, when the test assets are the 202 characteristic portfolios, the RMS alpha is 0.30%, which is larger than the corresponding value of the FF factors (0.19%). Also, augmenting these hedged five factors with PCA, PLS, or RRA factor is unable to improve the pricing power either.

In contrast to [Harvey, Liu, and Zhu \(2016\)](#) that there are too many factors in the asset pricing literature, we conclude this section that there are too few factors that are useful, and the future

Table 7 Performance of different factor models with pre-specified FF five factors

This table reports the root-mean-squared alphas and total adj- R^2 s of different factor models in explaining four sets of test assets. FF5 refers to the FF (2015) five-factor model, and FF5+PCA, FF5+PLS, and FF5+RRA refer to K -factor models that include FF5 and $(K - 5)$ PCA, PLS, and RRA factors, respectively. PCA factors are extracted from factor proxies excluding FF5; PLS and RRA refer to factors estimated with the 48 industry portfolios as basis assets. The sample period is 1974:01–2016:12.

$K =$	Root-mean-squared alpha (%)					Total adj- R^2 (%)				
	5	6	7	8	10	5	6	7	8	10
Panel A: 48 industry portfolios										
FF5	0.34	–	–	–	–	57.57	–	–	–	–
FF5+PCA	–	0.31	0.31	0.31	0.28	–	58.80	59.23	61.00	61.85
FF5+PLS	–	0.36	0.37	0.27	0.32	–	60.81	61.60	62.38	63.79
FF5+RRA	–	0.34	0.30	0.21	0.21	–	61.33	63.14	64.65	65.80
Panel B: 202 characteristic portfolios										
FF5	0.19	–	–	–	–	86.94	–	–	–	–
FF5+PCA	–	0.15	0.15	0.15	0.14	–	87.52	87.65	87.99	88.55
FF5+PLS	–	0.20	0.20	0.17	0.20	–	87.54	88.18	88.55	88.89
FF5+RRA	–	0.19	0.16	0.16	0.15	–	87.57	88.41	88.83	89.16
Panel C: All stocks										
FF5	3.08	–	–	–	–	14.70	–	–	–	–
FF5+PCA	–	3.22	3.28	3.43	3.61	–	15.26	15.42	15.79	16.24
FF5+PLS	–	3.12	3.18	3.39	3.87	–	15.29	15.88	16.18	16.59
FF5+RRA	–	3.18	3.18	3.30	3.41	–	15.18	15.75	16.01	16.60
Panel D: All-but-micro stocks										
FF5	2.06	–	–	–	–	28.39	–	–	–	–
FF5+PCA	–	2.13	2.16	2.20	2.27	–	29.10	29.42	30.28	31.01
FF5+PLS	–	2.10	2.11	2.18	2.47	–	29.66	30.34	30.90	31.77
FF5+RRA	–	2.12	2.13	2.21	2.27	–	29.75	30.86	31.48	32.28

research is to identify new factors that contain incremental information relative to the FF five factors.

4.6 Alternative PCA factors

In finding true factors that explain the returns of basis assets, one view is that the true factors are not functions of some given factor proxies, but rather latent functions of the basis assets. In line

with this view, we extract factors in (1) from the basis assets directly. Suppose the factors are linear functions of basis assets' returns,

$$f_{jt} = c_{j1}R_{1t} + \cdots + c_{jN}R_{Nt}, \quad j = 1, \dots, K. \quad (26)$$

With this assumption, the model can be written as

$$R_t = \alpha + BC'R_t + e_t, \quad (27)$$

where α is an N -vector of alphas, B is an $N \times K$ matrix of the factor loadings, C is an $N \times K$ matrix of factor weights on the basis assets, and e_t is an N -vector of the residuals.

There are many ways to obtain the estimates of C , and sequentially α and B by running regressions of R_t on $f_t = C'R_t$. A simple way is to minimize the expected mean-squared errors, $E(e_t'e_t)$. The solution is well known (Bai, 2003; Balvers and Stivers, 2018), and the extracted factors are

$$\hat{f}_t = \hat{C}'R_t, \quad (28)$$

where $\hat{C}$ is the first K eigenvectors of the sample covariance matrix of R_t :

$$\hat{\Sigma}_R = (R - \iota_T \hat{\mu}')'(R - \iota_T \hat{\mu}')/T. \quad (29)$$

That is, under the loss function of the expected mean-squared errors, the extracted factors are obtained from the first K principal components of the basis asset returns. Note that f_t is not unique because any rotation will give rise to new factors. However, α and BC' are unique.

Balvers and Stivers (2018) provide a novel approach to extract factors from returns under a pricing error restriction of the form $\alpha'\alpha = \eta\mu'\mu$ with η as a given constant. They solve the estimator almost analytically by minimizing the mean-squared error loss function. For the exact pricing relation $\alpha = 0$, we are able to provide a simple analytical expression. Specifically, imposing

a zero pricing error restriction, $\alpha = 0$, we have from equation (27) that

$$e_t = (I - BC')R_t. \quad (30)$$

Hence, minimizing the mean-squared residuals is to

$$\min_{B,C} E[e_t' e_t] \equiv \text{tr} [(I - BC')(I - BC')(\Sigma + \mu\mu')(I - BC')]. \quad (31)$$

Comparing this with equation (28), we have now the same objective function as before except that $\Sigma + \mu\mu'$ plays the role of the previous Σ . Hence, the solution can be analytically obtained, and the factors (with a zero pricing error restriction) are

$$\hat{f}_t = \hat{C}' R_t, \quad (32)$$

where $\hat{C}' = (\hat{\Sigma} + \hat{\mu}\hat{\mu}')^{-1/2} \hat{E}'$, with $\hat{E}'$, $N \times K$, as the K standardized eigenvectors of $\hat{\Sigma} + \hat{\mu}\hat{\mu}'$ corresponding to the K largest eigenvalues. By design, factors $\hat{f}_t$ is the best to fit the model or to explain the variation of the returns conditional on zero alphas.

Noting a salient feather that the basis assets usually do not have a mean zero, [Lettau and Pelger \(2020\)](#) propose a risk-premium PCA (RP-PCA) approach, which is equivalent to applying PCA to the matrix

$$\Sigma + (1 + \gamma)\mu\mu'. \quad (33)$$

where γ is a free parameter. Apparently, RP-PCA reduces to PCA when $\gamma = -1$ and to the zero mispricing case in equation (32) when $\gamma = 0$. Empirically, [Lettau and Pelger \(2020\)](#) find that the RP-PCA factors generate the highest Sharpe ratio when $\gamma = 10$.

The goal of our paper is to shrink the dimensionality of the factor space, and so we apply RRA to factor proxies to reduce a large number of factors to a few parsimonious ones. In contrast,

Table 8 Performance of alternative PCA factors

This table reports the root-mean-squared alphas and total adj- R^2 s of alternative PCA factors in pricing four sets of test assets, where the factors are directly extracted from the basis assets (i.e., 48 industry portfolios). BS-PCA and RP-PCA refer to the methods developed by [Balvers and Stivers \(2018\)](#) and [Lettau and Pelger \(2020\)](#), respectively. The sample period is 1974:1–2016:12.

$K =$	Root-mean-squared alpha (%)				Total adj- R^2 (%)			
	1	3	5	6	1	3	5	6
Panel A: 48 industry portfolios								
PCA	0.23	0.20	0.15	0.15	55.74	67.11	73.76	75.69
RP-PCA	0.23	0.18	0.15	0.15	55.73	67.02	73.74	75.67
BS-PCA	0.23	0.16	0.14	0.14	55.47	66.49	73.57	75.48
Panel B: 202 characteristic portfolios								
PCA	0.30	0.28	0.30	0.30	75.60	77.39	80.93	81.09
RP-PCA	0.29	0.28	0.31	0.31	75.44	78.18	81.01	81.16
BS-PCA	0.29	0.30	0.32	0.32	74.45	78.73	81.05	81.19
Panel C: All stocks								
PCA	2.49	2.60	2.65	2.71	9.51	11.08	13.51	13.86
RP-PCA	2.50	2.61	2.66	2.73	9.44	11.63	13.54	13.86
BS-PCA	2.51	2.65	2.70	2.79	9.13	11.87	13.52	13.78
Panel D: All-but-micro stocks								
PCA	1.73	1.74	1.82	1.81	21.07	24.68	28.77	29.31
RP-PCA	1.73	1.76	1.82	1.82	20.98	25.34	28.75	29.27
BS-PCA	1.75	1.77	1.82	1.82	20.53	25.06	28.66	29.15

[Balvers and Stivers \(2018\)](#) and [Lettau and Pelger \(2020\)](#) apply the PCA to extract factors from a given set of basis assets by imposing pricing error constraints. It is important to understand the difference between their approaches and ours. Their approaches and ours are entirely different. This is because they extract factors from the returns of basis assets, and do not use any information of factor proxies. In contrast, RRA extracts the most relevant factors from the set of factor proxies by shrinking the factor dimension. As a result, their approaches and ours cannot be compared directly as they apply to different objects.

However, there is an extreme situation in which the basis assets are the same as the factor proxies. In this case, RRA is degenerated and reduced to applying PCA to the basis assets. For brevity, we consider only the case with the 48 industry portfolios as the basis assets and report in

Table 8 the in-sample results of extracting factors with PCA, [Balvers and Stivers \(2018\)](#) PCA (dubbed as BS-PCA), and RP-PCA, respectively. The results make two observations (out-of-sample results display similar patterns, although slightly weaker, and are omitted for brevity). First, since the PCA factors are extracted from the 48 industry portfolios, they perform slightly better than the RRA factors extracted from the 70 factor proxies and much better than the FF factors. This is possible because the latter are broad factors. Indeed, when pricing the 202 characteristic portfolios, these PCA factors perform worse than the RRA factors. For example, the RMS alpha and total adj- R^2 of the PCA five factors for pricing the 202 characteristic portfolios are 0.32% and 81.05%, while the corresponding values with the RRA five factors are 0.27% and 84.60%. However, when pricing the individual stocks, the RRA and PCA factors perform similarly, although the RRA factors generate greater total adj- R^2 s. Noted that this has nothing to do with RRA as a valid statistical method designed to best extract factors from the factor proxies, but it speaks of the quality of the factor proxies. Consistent with early analysis, the result indicates that the existing factor proxies do not contain much information beyond the FF five factors in pricing individual stocks.

Second, both the recently proposed BS-PCA and RP-PCA perform similarly as PCA. This result is different from [Lettau and Pelger \(2020\)](#) who find that RP-PCA is able to identify weak factors and does a better job in explaining anomaly portfolios. One possible reason is that all the 48 industry portfolios follow a strong factor structure and are loading on all the latent factors. This is highly possible as the weakness of a factor depends on basis (or test) assets ([Giglio, Xiu, and Zhang, 2021](#)). Indeed, the 70 factor proxies have a canonical correlation at least 0.60 with each of the 48 industry portfolios, suggesting that they are likely strong and loaded by the 48 industry portfolios. In this sense, we do not face the weak factor issue in [Giglio, Xiu, and Zhang \(2021\)](#).⁶

⁶If some of the basis assets have a zero exposure on the factor proxies or some of the factor proxies are redundant, they would affect the extraction of true factors and consequently result in a failure in risk premium inference [see [Giglio, Kelly, and Xiu \(2021\)](#) and reference therein], which has been a long-standing issue in asset pricing. Recently, the literature has made some progress. For example, [Giglio, Xiu, and Zhang \(2021\)](#) argue that factor strength is not an inherent property of a factor, but depends on the selection of basis assets. As such, they suggest selecting basis assets with sufficient exposure on the factors of interest before conducting risk premium inference.

Table 9 Performance of LASSO selected factors

This table reports the root-mean-squared alphas and total adj- R^2 s of LASSO selected factors for pricing four sets of test assets. Panel A follows [Feng, Giglio, and Xiu \(2020\)](#) and selects factors from 70 factor proxies that best explain the average returns of the 48 industry portfolios. Panel B follows [Kozak, Nagel, and Santosh \(2020\)](#) and selects PCA factors that best represents the investment opportunity set spanned by the 70 factor proxies. CV refers to factors selected with cross-validation. The sample period is 1974:01–2016:12.

$K =$	Root-mean-squared alpha (%)				Total adj- R^2 (%)			
	3	5	6	CV	3	5	6	CV
Panel A: Feng, Giglio, and Xiu (2020) method								
48 industry portfolios	0.97	0.20	0.20	0.24	19.35	56.15	57.66	51.39
202 characteristic portfolios	1.14	0.41	0.41	0.32	36.40	80.09	80.83	73.31
All stocks	2.92	2.96	3.01	2.53	8.47	12.61	13.46	9.37
All-but-micro stocks	2.20	2.02	1.97	1.78	14.06	26.15	27.51	21.07
Panel B: Kozak, Nagel, and Santosh (2020) method								
48 industry portfolios	0.27	0.30	0.29	0.28	54.15	53.91	53.50	52.41
202 characteristic portfolios	0.29	0.28	0.27	0.25	75.69	75.48	75.28	74.44
All stocks	2.83	2.93	2.98	3.30	11.44	11.34	11.21	10.48
All-but-micro stocks	2.05	2.11	2.14	2.25	24.54	24.16	23.81	22.47

4.7 LASSO-based factors

In the literature, LASSO has been widely used to choose factors. This section considers LASSO with two different implementations. The first implementation is to select K factors from the 70 factor proxies to best explain the average returns of the 48 industry portfolios, where K is pre-specified or determined via cross-validation ([Feng, Giglio, and Xiu, 2020](#)). The second implementation is to select $K - 1$ principal components from the non-market factor proxies to best represents the investment opportunity set ([Kozak, Nagel, and Santosh, 2020](#)). Augmenting the $K - 1$ factors with the market factor generates a K -factor model.

Table 9 reports the results (as other tables, we focus on the in-sample results and ignore the out-of-sample ones as the latter are similar). With the first implementation in Panel A, a six-factor model yields the smallest RMS alpha and largest total adj- R^2 for pricing the 48 industry portfolio, whereas the cross-validation model select the market factor only. In comparison, as shown in Table

2, an RRA three-factor model can generate the same RMA alpha but higher total adj- R^2 (61.11% vs. 57.66%). When using the selected factors to price the other three sets of test assets, the pricing performance is always worse than the RRA factors. The reason is that the RRA factors are linear combinations of all the factor proxies and cover the LASSO factors as a special case.⁷

In Panel B, with the second implementation, a three-factor model performs the best in terms of RMS alpha and total adj- R^2 , and its performance is comparable with the RRR one-factor model. For example, the RMS alpha and total adj- R^2 for pricing the 48 industry portfolios are 0.27% and 54.15%, which are close to 0.23% and 54.60% with the RRA factors. However, including more factors will worsen, rather than improve, the pricing performance. In sum, RRA works better than LASSO in the context of this paper.

5 Conclusion

In this paper, we provide a reduced-rank approach (RRA) for shrinking factor dimension, a solution to dealing with the large number of factor proxies discovered by the empirical finance research. In contrast to other dimension reduction tools like PCA, PLS, and LASSO, RRA is designed to explain the cross section of expected stock returns and implemented analytically with asymptotic theory available on the properties of estimates and hypothesis testing. For example, to find five factors out of 70 proxies, there are millions possibilities in taking five out of 70 as well as taking five linear combinations out of the 70 ones, so that a direction verification of model fitting is computationally highly demanding if not impossible. RRA finds the best five linear combinations analytically that are the best in explaining the underlying basis assets by minimizing the mean-squared errors.

We apply RRA to 70 potential factor proxies, including the well known Fama-French five factors, momentum factor, [Pastor and Stambaugh \(2003\)](#) liquidity factor, [Hou, Xue, and Zhang](#)

⁷The LASSO factors perform similarly as the RRA factors in pricing the four sets of test assets when the 202 characteristic portfolios are used as basis assets. We report this result in the Online Appendix as our benchmark in the main text is the 48 industry portfolios as basis assets.

(2015) ROE factor, and 62 CAPM anomaly portfolios. We find that, while the 70 factor proxies improve the pricing performance at the portfolio level, they do not provide much new information at the stock level relative to the CAPM. Applications of PCA and PLS yield the same conclusion, though the latter two are less efficient in extracting factors from observable proxies. In addition, we extend RRA to include the FF five factors as pre-specified factors, and find that linear combinations of the remaining factor proxies do not improve much the performance.

Future research is to find more on the nature of mispricing and understand to what extent the pricing errors can be reduced by factor proxies. Nevertheless, the RRA methodology can be used to reduce the dimensionality of a factor space to find the minimum number of factors necessary.⁸ It will be of interest to apply RRA to both international equity markets and to other asset classes such as bonds and currencies.

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⁸Reinsel, Velu, and Chen (2022) recently extend the reduced-rank regression to allow N to be large relative to T , which can be potentially adapted in this paper to accommodate a large number of factors for any given T .

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Appendix

A Derivation of the GMM estimators

To prove the formulas, we write first the objective function as

$$Q = \text{tr}[W_1(R - \iota_T \alpha' - \Theta' G)' P_0(R - \iota_T \alpha' - \Theta' G)]/T^2. \quad (34)$$

Now we solve α conditional on other parameters. Let U be a $T \times N$ matrix of the residuals or the matrix obtained from stacking the U_t 's. Then it can be verified that

$$U' P_0 U = (R - \iota_T \hat{\alpha}' - \Theta' G)'(R - \iota_T \hat{\alpha}' - \Theta' G) + (\alpha' - \hat{\alpha}')' \iota_T' P_0 \iota_T (\alpha' - \hat{\alpha}'). \quad (35)$$

Hence, regardless of W_1 , Q is minimized at $\alpha = \hat{\alpha}$. Consequently, we only need to minimize

$$Q^* = \text{tr}[W_1(R - \Theta' G)' P(R - \Theta' G)]/T^2. \quad (36)$$

The rest of the proof follows [Zhou \(1994\)](#). Q.E.D.

B D_T representation

For the reader's application convenience, this appendix provides an explicit expression for D_T for both computing the GMM test and for checking the optimality of the estimators with $D_T' W_T h_T$ being zero.

Recall that the regression system is:

$$\begin{aligned} R_t &= \alpha + \beta' f_t + U_t, \quad \beta : K \times N \\ &= \alpha + \Theta' g_t + U_t, \quad \Theta : L \times N \\ f_t &= \Phi' g_t, \quad \Phi : L \times K, \quad \Theta = \Phi \beta. \end{aligned}$$

To make the solution unique, Φ and β can be represented as

$$\Phi = \begin{pmatrix} \phi_{11} & \phi_{12} & \cdots & \phi_{1K} \\ \phi_{21} & \phi_{22} & \cdots & \phi_{2K} \\ \cdots & \cdots & \cdots & \cdots \\ \phi_{L1} & \phi_{L2} & \cdots & \phi_{LK} \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 0 & \cdots & 0 & \beta_{1,K+1} & \cdots & \beta_{1N} \\ 0 & 1 & \cdots & 0 & \beta_{2,K+1} & \cdots & \beta_{2N} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 1 & \beta_{K,K+1} & \cdots & \beta_{KN} \end{pmatrix} = (I_K, \beta_2).$$

The parameters to be estimated are

$$\begin{aligned}\theta &= (\alpha', \text{vec}(\Phi)', \text{vec}(\beta_2)')' \\ &= (\alpha_1, \dots, \alpha_N, \phi_{11}, \dots, \phi_{L1}, \dots, \phi_{1K}, \dots, \phi_{LK}, \beta_{1,K+1}, \dots, \beta_{K,K+1}, \dots, \beta_{1N}, \dots, \beta_{KN}).\end{aligned}$$

Let $h_t = U_t(\alpha, \Phi, \beta_2) \otimes Z_t$ with $U_t = R_t - \alpha - (\Phi\beta)'g_t = R_t - \alpha - \begin{pmatrix} \Phi'g_t \\ \beta_2'\Phi'g_t \end{pmatrix}$ and $Z_t = (1, g_t')'$.

Then

$$\Phi'g_t = \begin{pmatrix} \sum_{i=1}^L \phi_{i1}g_{it} \\ \sum_{i=1}^L \phi_{i2}g_{it} \\ \dots \\ \sum_{i=1}^L \phi_{iK}g_{it} \end{pmatrix} \quad \text{and} \quad \beta_2'\Phi'g_t = \begin{pmatrix} \sum_{j=1}^K [\beta_{j,K+1} \sum_{i=1}^L \phi_{ij}g_{it}] \\ \sum_{j=1}^K [\beta_{j,K+2} \sum_{i=1}^L \phi_{ij}g_{it}] \\ \dots \\ \sum_{j=1}^K [\beta_{j,N} \sum_{i=1}^L \phi_{ij}g_{it}] \end{pmatrix}.$$

Since $\frac{\partial U_t}{\partial \alpha} = -I_N$,

$$\frac{\partial \Phi'g_t}{\partial \text{vec}(\Phi)} = \begin{pmatrix} g_{1t} & \dots & g_{Lt} & \cdot & \dots & \cdot & 0 & \dots & 0 \\ \vdots & \dots & \vdots & \vdots & \dots & \vdots & \vdots & \dots & \vdots \\ 0 & \dots & 0 & \cdot & \dots & \cdot & g_{1t} & \dots & g_{Lt} \end{pmatrix} = I_K \otimes g_t',$$

$$\frac{\partial \beta_2'\Phi'g_t}{\partial \text{vec}(\Phi)} = \begin{pmatrix} \beta_{1,K+1}g_{1t} & \dots & \beta_{1,K+1}g_{Lt} & \cdot & \dots & \cdot & \beta_{K,K+1}g_{1t} & \dots & \beta_{K,K+1}g_{Lt} \\ \vdots & \dots & \vdots & \vdots & \dots & \vdots & \vdots & \dots & \vdots \\ \beta_{1,N}g_{1t} & \dots & \beta_{1,N}g_{Lt} & \cdot & \dots & \cdot & \beta_{K,N}g_{1t} & \dots & \beta_{K,N}g_{Lt} \end{pmatrix} = \beta_2' \otimes g_t',$$

and

$$\frac{\partial \beta_2'\Phi'g_t}{\partial \text{vec}(\beta_2)} = \begin{pmatrix} \sum_{i=1}^L \phi_{i1}g_{it} & \dots & \sum_{i=1}^L \phi_{iK}g_{it} & \cdot & \dots & \cdot & 0 & \dots & 0 \\ \vdots & \dots & \vdots & \vdots & \dots & \vdots & \vdots & \dots & \vdots \\ 0 & \dots & 0 & \cdot & \dots & \cdot & \sum_{i=1}^L \phi_{i1}g_{it} & \dots & \sum_{i=1}^L \phi_{iK}g_{it} \end{pmatrix} = I_{N-K} \otimes g_t'\Phi,$$

the first-order derivative of h_t with respect to θ is

$$D_t = \frac{\partial h_t}{\partial \theta} = \frac{\partial U_t}{\partial \theta} \otimes Z_t = - \begin{pmatrix} I_K & 0 & I_K \otimes g_t' & 0 \\ 0 & I_{N-K} & \beta_2' \otimes g_t' & I_{N-K} \otimes g_t'\Phi \end{pmatrix} \otimes Z_t.$$

Thus, $D_T = \frac{1}{T} \sum_{t=1}^T D_t$. Q.E.D.